1 Abstract Definition of Probability

Probability is the area of mathematics concerned with describing uncertain or random events. We will develop a mathematical framework that will allow us to quantify uncertainty in a principled way.

1.1 Probability Space

A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ consists of three parts: Ω denotes the set of outcomes, $\mathcal{F}$ the events we can assign probability to, and $\mathbb{P}$ the probability of each event. This forms the foundation of probability theory.

The collection of possible outcomes is called the probability space, from which we can associate possible events and their respective probabilities.

Definition 1. A nonempty set Ω is called a **probability space**. The elements of Ω are called outcomes.

Unless Ω is countable, the space of subsets is often too rich to understand. However, if we restrict ourselves to a large, yet natural collection of subsets, we will still retain enough sets to build a rich theory of probability from essential ideas in analysis.

Definition 2. A collection of subsets $\mathscr{F}$ of Ω is called a **σ-algebra** on Ω if

1. $\Omega \in \mathscr{F}$
2. Closed under complements: if $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$
3. Closed under countable unions: if $A_1, A_2, \ldots \in \mathscr{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$.

The elements of $\mathcal{F}$ are often called **events**.

Remark 1. When $\Omega = \mathbb{R}^n$, it is typically not possible to take $\mathcal{F}$ to be the power set consisting of all subsets of $\mathbb{R}^n$. Some sets in $\mathbb{R}^n$ are very weird, and need to be excluded. The classical choice of σ-algebra on $\mathbb{R}^n$ is the **Borel σ-algebra**

$$\mathscr{B}(\mathbb{R}^n),$$

which is defined as the smallest σ-algebra that contains all open cubes

$$(a_1, b_1) \times \cdots \times (a_n, b_n) \qquad \text{with } a_i < b_i$$

It is quite difficult to construct sets that are not Borel sets, so $\mathscr{B}(\mathbb{R}^n)$ contains all "reasonable" sets.

The main consequence of restricting our study to σ-algebras is the existence of a way to measure the likelihood of events.

Definition 3. A function $\mathbb{P}$ is a **probability measure** on the σ-algebra $\mathcal{F}$ if

1. $\mathbb{P}(\Omega) = 1$;
2. $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$;
3. Countable Additivity: if $A_1, A_2, \ldots \in \mathscr{F}$ are disjoint ($A_i \cap A_j = \emptyset$ for all $i \neq j$), then

$$\mathbb{P}\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

1.1.1 Discrete Probability Spaces

If Ω is countable, then we can always assign a probability to every outcome. Thus a probability measure is completely determined by the probabilities of each individual outcome.

Corollary 1

Let $\Omega = \{\omega_1, \omega_2, \dots\}$ and let $A \subseteq \Omega$ be an event. Then

$$\mathbb{P}(A) = \sum_{\omega_i \in A} \mathbb{P}(\omega_i).$$

Remark 2. On the other hand, if Ω is uncountable, then it is impossible to assign a probability to every outcome, so it is necessary to define probabilities on the set of measurable events $\mathcal{F}$.

A special case of a discrete probability space with finitely many elements $\Omega = \{\omega_1, \dots, \omega_n\}$ is the one with *equally likely outcomes*. In this case, $\mathbb{P}(\omega_1) = \mathbb{P}(\omega_2) = \cdots = \mathbb{P}(\omega_n)$ so the normalization requirement implies that for any index $k \in \{1, \dots, n\}$

$$1 = \mathbb{P}(\Omega) = \sum_{\omega_i \in \Omega} \mathbb{P}(\omega_i) = |\Omega| \, \mathbb{P}(\omega_k) \implies \mathbb{P}(\omega_k) = \frac{1}{|\Omega|}.$$

This is often the most naive definition of a probability space.

Definition 4 (Uniform Probability Measure)**.** Let Ω be finite. If all outcomes are equally likely, then the associated probability measure $\mathbb{P}$ on Ω is called the *uniform probability measure* and

$$\mathbb{P}(A) = \sum_{\omega_i \in A} \mathbb{P}(\omega_i) = \sum_{\omega_i \in A} \frac{1}{|\Omega|} = \frac{|A|}{|\Omega|}.$$

This means that for uniform probability measures, we can simply count the number of ways an event can happen to compute the probabilities on finite probability spaces. However, we will encounter much richer probability spaces throughout this course, since outcomes might not always be equally likely. Understanding these probabilities will require tools beyond counting.

1.2 Basic Calculations with Probabilities

From these three properties, we can recover all the natural properties a probability should satisfy.

Proposition 1 (*Basic Properties of a Probability Measure*)

Any probability measure $\mathbb{P}$ satisfies the following

1. $\mathbb{P}(A) \in [0, 1]$,
2. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$,
3. *Monotonicity: If* $A \subseteq B$, *then* $\mathbb{P}(A) \le \mathbb{P}(B)$,
4. *Inclusion–Exclusion Principle:* $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,
5. *Union Bound:* $\mathbb{P}(A \cup B) \le \mathbb{P}(A) + \mathbb{P}(B)$.

More advanced formulas to compute more complicated events will be introduced in the next section.

1.3 Example Problems

Problem 1.1. Suppose two six sided dice are rolled, and the number of dots facing up on each die is recorded.

1. Write down the probability space Ω.
2. Write down, as a set, the event $A =$ "The sum of the dots is 7".
3. Write down, as a set, the event B^c, where $B =$ "The sum of the numbers is at least 4".
4. Write down, as a set, the events $A \cap B^c$ and $A \cup B^c$.

Solution 1.1.

1. The sample space for a pair of dice is the a pair of the outcomes of each die roll
$$\Omega = \{1, \dots, 6\} \times \{1, \dots, 6\} = \{(x, y) : x, y \in \{1, 2, \dots, 6\}\}.$$
2. We can simply write down all the combinations
$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$
3. If $B = \{\text{sum is at least } 4\}$ then $B^c = \{\text{sum is at most } 3\}$, so
$$B^c = \{(1, 1), (1, 2), (2, 1)\}.$$
4. Since it is impossible for the sum of dots to be 7 and at most 3 at the same time, $A \cap B^c = \emptyset$. All the possibilities the sum of dots is 7 or at most 3 is
$$A \cup B^c = \{\underbrace{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}_{A}, \underbrace{(1, 1), (1, 2), (2, 1)}_{B^c}\}.$$

Problem 1.2. For the following experiments, describe a possible probability space Ω.

1. Roll a die.
2. Number of coin-flips until heads occurs.
3. Waiting time in minutes (with infinite precision, e.g., $0.238\overline{445}$ minutes) until a task is complete.

Solution 1.2.

1. There are many ways we can record the outcome of a die such that no elements can occur at the same time $\Omega = \{1, 2, 3, 4, 5, 6\}$ or $\Omega = \{\text{even}, \text{odd}\}$. The choice of the best probability space will depend on the application in mind, but usually the coarsest choice is the most powerful.
2. There is only one natural choice here $\Omega = \{1, 2, 3, \dots\} = \mathbb{N}$.
3. There is only one natural choice here $\Omega = [0, \infty) = \{x \in \mathbb{R} : x \geq 0\}$.

Problem 1.3. Suppose that two fair six sided die are rolled.

1. What is the probability that the dots on each die match?

2. What is the probability that the dots sum to 7?

3. What is the probability that the dots do not sum to 7?

4. What is the probability that the dots match and sum to 7?

Solution 1.3. The probability is uniform over the probability space $\Omega = \{1, \dots, 6\}^2$. All outcomes are equally likely, so for an event A,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{36}.$$

1. The event is $A = \{(1,1), (2,2), \dots, (6,6)\}$ with $|A| = 6$, so $\mathbb{P}(A) = 6/36 = 1/6$.

2. The event is $B = \{(1,6), (2,5), \dots, (6,1)\}$ with $|B| = 6$, so $\mathbb{P}(B) = 6/36 = 1/6$.

3. The event is B^c with $|B^c| = |\Omega| - |B| = 30$ elements, hence $\mathbb{P}(B^c) = 30/36 = 5/6 = 1 - \mathbb{P}(B)$.

4. 7 is an odd number so it is impossible for the dots to match. Therefore, $\mathbb{P}(\emptyset) = 0$.

Problem 1.4. Supposes that $\mathbb{P}(A) = \frac{1}{3}$ and $\mathbb{P}(B^c) = \frac{1}{4}$, can A and B be disjoint?

Solution 1.4. We have

$$\mathbb{P}(B) = 1 - \mathbb{P}(B^c) = \frac{3}{4}.$$

By the inclusion exclusion property, we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Suppose that A and B are disjoint, i.e. $\mathbb{P}(A \cap B) = 0$, then we must have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) = \frac{1}{3} + \frac{3}{4} = \frac{13}{12} > 1$$

which contradicts the fact that probabilities are between 0 and 1. Thus, A and B cannot be disjoint.

Problem 1.5. Let $\mathcal{F}$ be a σ-algebra. Show that $A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}, A \cup B \in \mathcal{F}$.

Solution 1.5. Since $\mathcal{F}$ is closed under countable unions we have closure under finite unions $A \cup B \in \mathcal{F}$ (by taking $A_3 = A_4 = \cdots = B$).

To see that $A \cap B \in \mathcal{F}$, notice that $A^c, B^c \in \mathcal{F}$ is so $A^c \cup B^c \in \mathcal{F}$ by above. By taking complements we have that $(A^c \cup B^c)^c = A \cap B \in \mathcal{F}$ using De Morgan's identity.

1.4 Proofs of Key Results

Problem 1.6. (Proposition 1) Show the *monotonicity* property of probability,

$$\text{if } A \subseteq B \text{ then } \mathbb{P}(A) \leq \mathbb{P}(B).$$

Solution 1.6. This follows directly from the axioms. If $A \subseteq B$, then

$$B = (A \cap B) \cup (A \setminus B) = A \cup (A \setminus B)$$

and the sets A and $A \setminus B$ are disjoint. Therefore, by countable additivity,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(A \setminus B) \geq \mathbb{P}(A)$$

since $\mathbb{P}(A \setminus B) \geq 0$ by the non-negativity property.

Problem 1.7. (Proposition 1) Show that the axiomatic definition of a probability implies that

$$0 \leq \mathbb{P}(A) \leq 1$$

for any event A.

Solution 1.7. By the monontonicity property, since $A \subseteq \Omega$,

$$\mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1$$

by the normalization property. On the other hand, but non-negativity, we have that $\mathbb{P}(A) \geq 0$.

Problem 1.8. (Proposition 1) Show that the axiomatic definition of a probability implies that

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$$

for any event A.

Solution 1.8. Notice that $A \cup A^c = \Omega$ and A and A^c are disjoint. From finite additivity, we conclude that

$$\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(\Omega) = 1 \implies \mathbb{P}(A) = 1 - \mathbb{P}(A^c).$$

Problem 1.9. (Proposition 1) Show that the axiomatic definition of a probability implies that

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

for any event A and B.

Solution 1.9. Notice that A and $(B \setminus A)$ are disjoint events such that $A \cup (B \setminus A) = A \cup B$, so by countable additvity

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$$

Next, notice that $B \setminus A$ and $A \cap B$ are exclusive and $(A \cap B) \cup (B \setminus A) = A \cap B$,

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) \implies \mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

which implies our result.

Problem 1.10. (Proposition 1) Show that the axiomatic definition of a probability implies that

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B),$$

for any event A and B.

Solution 1.10. By the inclusion–exclusion property,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$$

since $\mathbb{P}(A \cap B) \geq 0$. Notice that this implies that the union bound is sharp and is attained when A and B are disjoint sets, which is implied by countable additivity.

2 Exclusivity and Independence

2.1 Conditional Probability

Probabilities can change given more information. When we are given new information, we can restrict our sample space to outcomes that are consistent with this new information and update the probabilities in the following way.

Definition 5 (Conditional Probability)**.** The *conditional probability* of A given B is, provided that $\mathbb{P}(B) > 0$, is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Remark 3. In general, $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ do not give you the same value, so the order matters.

Essentially, if we are given that B occurred, then we restrict the sample space Ω to the new sample space of possible outcomes B and recompute the probabilities. The new events are now $A \cap B$ (since events have to be subsets of B) and the normalization by $\mathbb{P}(B)$ ensures that conditional probabilities are probabilities (since the probability of B given B must be 1). Conditional probabilities are probabilities, they satisfy many of the same properties:

1. $0 \leq \mathbb{P}(A \mid B) \leq 1$
2. If A_1 and A_2 are disjoint, then $\mathbb{P}(A_1 \cup A_2 \mid B) = \mathbb{P}(A_1 \mid B) + \mathbb{P}(A_2 \mid B)$
3. $\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B)$
4. $\mathbb{P}(\Omega \mid B) = 1 = \mathbb{P}(B \mid B)$

2.2 Mutually Independent and Mutually Exclusive Events

We extend the concept of the sum and product rule from counting to probabilities that are not necessarily uniform.

2.2.1 (Mutually) Exclusive Events

Recall that two events are *mutually exclusive* if they cannot happen at the same time.

Definition 6 (Exclusive Events)**.** Two events A and B are *exclusive* if $A \cap B = \emptyset$. A sequence of events $A_1, A_2, ..., A_n$ are said to be *mutually exclusive* if

$$A_{i_1} \cap A_{i_2} = \emptyset,$$

for all possible $i_1 \neq i_2$.

Remark 4. Mutually exclusive events satisfy the analogue of the "sum" rule. If $A_1, A_2, \dots, A_n$ are mutually exclusive events, then

$$\mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

This property is very useful because it mirrors how volumes are computed, in the sense that the volume of non-overlapping objects is equal to the sum of the volumes of individual objects.

2.2.2 (Mutually) Independent Events

Informally, two events are *independent* if they do not have an influence on each other.

Definition 7 (Independent Events I)**.** Two events A and B are independent, if

$$\mathbb{P}(A \mid B) = \mathbb{P}(A),$$

provided $\mathbb{P}(B) > 0$. (Or equivalently, $\mathbb{P}(B \mid A) = \mathbb{P}(B)$ provided that $\mathbb{P}(A) > 0$.)

From the definition of conditional probability, we arrive at the following equivalent notion.

Definition 8 (Independent Events II)**.** Two events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

A sequence of events $A_1, A_2, \ldots, A_n$ are said to be *mutually independent* if

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \mathbb{P}(A_{i_1})\,\mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_k}),$$

for all possible $1 \leq k \leq n$ and $1 \leq i_1 < \ldots < i_k \leq n$. Events that are *not independent* are *dependent*.

Remark 5. Mutually independent events satisfy the analogue of the "product" rule. If $A_1, A_2, \ldots, A_n$ are mutually independent events, then

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \prod_{i=1}^{n} \mathbb{P}(A_i).$$

This property is very useful because it mirrors how volumes are computed, in the sense that the volume of a high dimensional objects is equal to the product of it's one dimensional side lengths (see Problem 2.8).

2.2.3 Mutually Exclusive vs Mutually Independent

It is important to not confuse mutually independent events with mutually exclusive events. They are actually completely "opposite" features, since if A and B are independent then the occurrence of B does not affect the occurrence of A

$$A, B \text{ independent} \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B) \iff \mathbb{P}(A \mid B) = \mathbb{P}(A)$$

but if A and B are mutually exclusive, then the occurrence of B means that A cannot occur

$$A, B \text{ mutually exclusive} \implies \mathbb{P}(A \cap B) = 0 \iff \mathbb{P}(A \mid B) = 0.$$

In fact, events A and B cannot be both mutually exclusive and independent (unless one of the events has probability zero (see Problem 2.9)).

2.3 Advanced Calculations with Probabilities

2.3.1 Inclusion Exclusion Principle:

We can generalize the sum rule to events that are not mutually exclusive.

Theorem 1 (*Inclusion–Exclusion Principle*)

For any events, A and B

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

General Form: *For arbitrary events* $A_1, A_2, \dots, A_n$ *with* $n \geq 1$,

$$\begin{aligned}\mathbb{P}(\cup_{i=1}^n A_i) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) \\ - \sum_{i<j<k<l} \mathbb{P}(A_i \cap A_j \cap A_k \cap A_l) + \cdots + (-1)^{n-1}\, \mathbb{P}(\cap_{i=1}^n A_i).\end{aligned}$$

The inclusion exclusion principle is tricky to compute when n is large, so we sometimes use the union bound to get an inequality of the union of events.

Corollary 2 (*Union Bound*)

For any events A *and* B,

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

General Form: *For arbitrary events* $A_1, A_2, \dots, A_n$ *with* $n \geq 2$,

$$\mathbb{P}(\cup_{i=1}^n A_i) = \mathbb{P}(A_1) + \cdots + \mathbb{P}(A_n) = \sum_i \mathbb{P}(A_i).$$

2.3.2 Chain Rule

We can generalize the multiplication rule to events that are not mutually independent.

Theorem 2 (*Chain Rule*)

For any events A *and* B,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\, \mathbb{P}(B) = \mathbb{P}(B \mid A)\, \mathbb{P}(A).$$

General Form: *For arbitrary events* $A_1, A_2, \dots, A_n$ *with* $n \geq 2$,

$$\begin{aligned}\mathbb{P}(\cap_{i=1}^n A_i) &= \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 \mid A_1) \cdot \mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \\ &= \prod_{k=1}^n \mathbb{P}(A_k \mid \cap_{j=1}^{k-1} A_j).\end{aligned}$$

2.3.3 Law of Total Probability

It is often easier to compute probabilities of events by breaking into cases.

Definition 9. A sequence of sets $B_1, B_2, ..., B_n$ are said to *partition* the sample space Ω if $B_i \cap B_j = \emptyset$ for all $i \neq j$, and $\cup_{j=1}^n B_j = \Omega$.

Essentially, breaking into the "cases" $B_1, \dots, B_k$ is valid whenever they form a partition because the condition $B_i \cap B_j = \emptyset$ ensures that our cases are distinct (so we don't overcount) and the $\cup_{j=1}^n B_j = \Omega$ condition ensures that we have not missed any cases (so we don't undercount). The simplest partition of Ω is B and B^c, which essentially breaks into the cases B happens and B does not happen.

Theorem 3 (*Law of total probability*)

For any events A *and* B,

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \mathbb{P}(A \mid B)\, \mathbb{P}(B) + \mathbb{P}(A \mid B^c)\, \mathbb{P}(B^c)$$

General Form: *Suppose that $B_1, B_2, ..., B_n$ partition of Ω. Then for any event A,*

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_2) + \cdots + \mathbb{P}(A \cap B_n) \\ &= \mathbb{P}(A \mid B_1)\, \mathbb{P}(B_1) + \mathbb{P}(A \mid B_2) P(B_2) + \cdots + \mathbb{P}(A \mid B_n)\, \mathbb{P}(B_n).\end{aligned}$$

2.3.4 Bayes' Theorem

We can "flip" the terms in event and conditioning event using Bayes Theorem.

Theorem 4 (*Bayes' Theorem*)

For any events A and B,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\, \mathbb{P}(B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B)\, \mathbb{P}(B)}{\mathbb{P}(A \mid B)\, \mathbb{P}(B) + \mathbb{P}(A \mid B^c)\, \mathbb{P}(B^c)}$$

General Form: *Suppose that $B_1, B_2, ..., B_n$ partition S. Then for any event A,*

$$\mathbb{P}(B_i \mid A) = \frac{\mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i)}{\sum_{j=1}^n \mathbb{P}(A \mid B_j)\, \mathbb{P}(B_j)}.$$

In Bayesian statistics, we often call $\mathbb{P}(B)$ the *prior* and $\mathbb{P}(B \mid A)$ the *posterior*. Bayes' theorem gives us a rule for how our prior probability adapts given more information A.

2.4 Example Problems

Problem 2.1. You roll a die and your friend looks at what number you rolled.

- What's the probability that it's a 6?
- What's the probability that it's a 6 given that your friend tells you that it's an even number?
- What's the probability that it's a 6 given that your friend tells you that it's an odd number?

Solution 2.1.

1. Since all outcomes are equally likely,

$$\mathbb{P}(\text{roll } 6) = \frac{1}{6}.$$

Remark 6. This answer makes sense since in the absence of more information, the sample space of possible outcomes is still $\{1, 2, 3, 4, 5, 6\}$, and the probability is uniform over this set.

2. By definition

$$\mathbb{P}(\text{roll } 6 \mid \text{even}) = \frac{\mathbb{P}(\text{roll } 6, \text{even})}{\mathbb{P}(\text{even})} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}.$$

Remark 7. This answer makes sense since if we know that the roll was even, the restricted sample space of possible outcomes is now $\{2, 4, 6\}$, and we have no reason to expect that any of these possible outcomes is more likely than the other, so the probability is now $1/3$.

3. By definition

$$\mathbb{P}(\text{roll } 6 \mid \text{odd}) = \frac{\mathbb{P}(\text{roll } 6, \text{odd})}{\mathbb{P}(\text{odd})} = \frac{0}{\frac{1}{2}} = 0.$$

Remark 8. This answer makes sense since if we know that the roll was odd, the restricted sample space of possible outcomes is now $\{1, 3, 5\}$. Since 6 is no longer a possible outcome, the probability is now 0.

Problem 2.2. Consider rolling two fair six sided dice, and let

$$A = \{\text{the sum is } 10\}, \quad B = \{\text{the first die is a } 6\}, \quad C = \{\text{the sum is } 7\}.$$

1. Compute $\mathbb{P}(A \mid B)$
2. Compute $\mathbb{P}(B \mid A)$
3. Compute $\mathbb{P}(A \mid C)$
4. Compute $\mathbb{P}(C \mid B)$

Solution 2.2. This problem is a direct application of the definition of conditional probabilities,

1. Since $\mathbb{P}(A \cap B) = \mathbb{P}((6, 4)) = 1/36$ we find $\mathbb{P}(A \mid B) = \frac{1/36}{1/6} = \frac{1}{6}$.
2. Similarly, $\mathbb{P}(B \mid A) = \frac{1/36}{1/12} = \frac{1}{3}$.
3. Since $A \cap C = \emptyset$, we have $\mathbb{P}(A \cap C) = 0$ and $\mathbb{P}(A \mid C) = \frac{0}{\mathbb{P}(C)} = 0$.
4. Since $\mathbb{P}(C \cap B) = \mathbb{P}((6, 1)) = 1/36$ we find $\mathbb{P}(C \mid B) = \frac{1/36}{1/6} = \frac{1}{6}$.

Problem 2.3. A survey company found that 63% of Canadians support Canadian Tire and 80% support Tim Hortons, while 51% of Canadians support both Tim Hortons and Canadian Tire. Let A denote the event that an individual supports Canadian Tire and B the event that an individual supports Tim Hortons.

1. Suppose a Canadian is selected at random, what is the probability that the individual supports Canadian Tire or Tim Hortons?
2. Suppose a Canadian is selected at random, what is the probability that the individual supports Tim Hortons but not Canadian Tire?

Solution 2.3.

Part 1: We are given that $\mathbb{P}(A) = 0.63$, $\mathbb{P}(B) = 0.8$ and $\mathbb{P}(A \cap B) = 0.51$ and we are asked to find $\mathbb{P}(A \cup B)$. By the inclusion exclusion principle,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.63 + 0.8 - 0.51 = 0.92.$$

Part 2: We are given that $\mathbb{P}(A) = 0.63$, $\mathbb{P}(B) = 0.8$ and $\mathbb{P}(A \cap B) = 0.51$ and we are asked to find $\mathbb{P}(A^c \cap B)$. By the law of total probability,

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B) \implies \mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.8 - 0.51 = 0.29.$$

Problem 2.4. The probability a randomly selected male is colour-blind is 0.05, whereas the probability a female is colour-blind is only 0.0025. If the population is 45% male, what is the fraction that is colour-blind?

Solution 2.4. Let

$C =$ "randomly selected person is colorblind" and $M =$ "randomly selected person is male."

We are given

$$\mathbb{P}(C \mid M) = 0.05;\ \ \mathbb{P}(C \mid M^c) = 0.0025;\ \ \mathbb{P}(M) = 0.45;\ \ \mathbb{P}(M^c) = 0.55$$

and want to find $\mathbb{P}(C)$. The law of total probability implies that

$$\mathbb{P}(C) = \mathbb{P}(C \mid M)P(M) + \mathbb{P}(C \mid M^c)P(M^c) = 0.05 \cdot 0.45 + 0.0025 \cdot 0.55 = 0.023875.$$

Problem 2.5. Justin has 12 red hats and 7 green hats. On Monday, Justin picks hats for his Monday and Wednesday lectures. He picks one hat at random (for the Monday lecture) and then, without replacement, another hat (for the Wednesday lecture).

1. What is the probability that both hats are red?
2. What is the probability that Justin wears a red hat on Wednesday?

Solution 2.5. The second hat I pick depends on the outcome of the first hat, so we should use conditional probabilities to compute our answers.

Part 1. Let

$$R_1 = \text{"first pair is red"} \quad \text{and } R_2 = \text{"second pair is red"}.$$

By the chain rule,

$$\begin{aligned}\mathbb{P}(R_1 \cap R_2) &= \mathbb{P}(R_1) \cdot \mathbb{P}(R_2 \mid R_1) \\ &= \frac{12}{19} \cdot \frac{11}{18} = \frac{22}{57} \approx 0.386\end{aligned}$$

Part 2: By the law of total probability,

$$\begin{aligned}\mathbb{P}(R_2) &= \mathbb{P}(R_2 \cap R_1) + \mathbb{P}(R_2 \cap R_1^c) \\ &= \mathbb{P}(R_2 \mid R_1)\,\mathbb{P}(R_1) + \mathbb{P}(R_2 \mid R_1^c)\,\mathbb{P}(R_1^c) \\ &= \frac{12}{19} \cdot \frac{11}{18} + \frac{7}{19} \cdot \frac{12}{18} \\ &= \frac{12}{19} \approx 0.632\end{aligned}$$

Problem 2.6. If I pick a red hat, there is a 20% chance somebody will laugh. If I pick a green hat, there is a 10% chance somebody will laugh. 70% of the time I choose a red hat, and 30% of the time a green hat. Given that somebody laughed, what is the probability that I picked a green hat?

Solution 2.6. Denote by

$$L = \text{“somebody laughed”} \quad \text{and} \quad G = \text{“I wore a green hat”}$$

We are given that $\mathbb{P}(L \mid R) = 0.2$, $\mathbb{P}(L \mid G) = 0.1$, $\mathbb{P}(R) = 0.7$ and $\mathbb{P}(G) = 0.3$. We need to compute the “flipped” conditional probability $\mathbb{P}(G \mid L)$. By Bayes' rule

$$\mathbb{P}(G \mid L) = \frac{\mathbb{P}(L \mid G)\,\mathbb{P}(G)}{\mathbb{P}(L \mid R)\,\mathbb{P}(R) + \mathbb{P}(L \mid G)\,\mathbb{P}(G)} = \frac{0.1 \cdot 0.3}{0.2 \cdot 0.7 + 0.1 \cdot 0.3} = 0.1765.$$

Problem 2.7. You have n identical looking keys on a chain, and one opens your office door. Suppose you try the keys in a random order (without selecting an incorrect key more than once). What is the probability the k'th key opens the door?

Solution 2.7. Let A_i denote the event that the ith key opens the door. We want to find

$$\mathbb{P}(A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c \cap A_k).$$

This problem is computable using the chain rule,

$$\begin{aligned}\mathbb{P}(A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c \cap A_k) &= \mathbb{P}(A_1^c) \cdot \mathbb{P}(A_2^c \mid A_1^c) \dots \mathbb{P}(A_k \mid (A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c) \\ &= \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdots \frac{n-k+1}{n-k+2} \cdot \frac{1}{n-k+1} = \frac{1}{n}.\end{aligned}$$

We used the fact that

1. $\mathbb{P}(A_1^c) = \frac{n-1}{n}$ because there are $n-1$ wrong keys out of n keys in total,
2. $\mathbb{P}(A_2^c \mid A_1^c) = \frac{n-2}{n-1}$ since there are $n-2$ wrong keys out of the remaining $n-1$ keys, because we know that the first key we tried failed
3. $\mathbb{P}(A_j^c \mid A_1^c \cap \cdots \cap A_{j-1}^c) = \frac{n-j}{n-j+1}$ by inductive reasoning
4. and lastly, $\mathbb{P}(A_k \mid (A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c) = \frac{1}{n-k+1}$ since there is one correct key out of the remaining $n-(k-1)$ keys (because we tried $k-1$ keys that failed to open the lock).

Alternative Solution I: The sample space for this problem are the ordered tuples of length n. If we know that the right key is used on the kth try, there are $(n-1)!$ choices for the other keys. So

$$\mathbb{P}(A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c \cap A_k) = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Alternative Solution II: We can take the sample space to be the location of the right key out of n possible locations. Therefore, $\Omega = [n]$ and the location of the right key is uniform over this sample space by symmetry, so

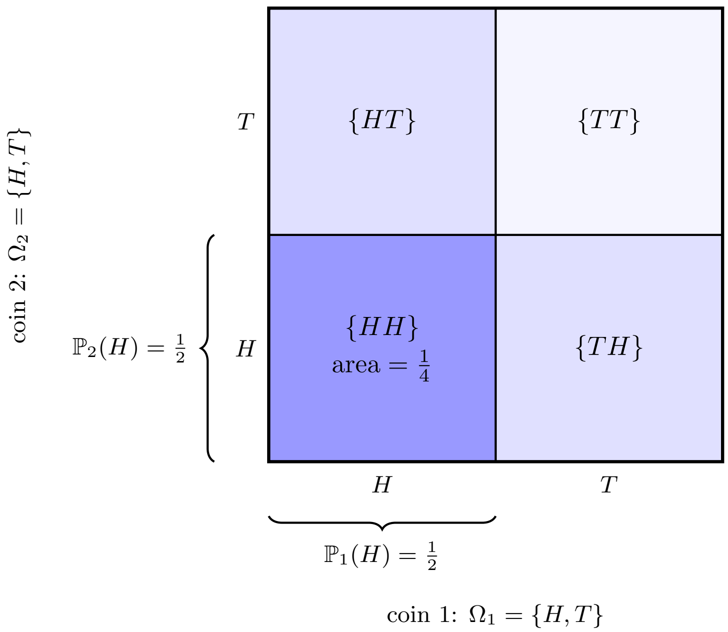
$$\mathbb{P}(A_1^c \cap A_2^c \cap \cdots \cap A_{k-1}^c \cap A_k) = \frac{1}{|\Omega|} = \frac{1}{n}.$$

Problem 2.8.

1. We flip a coin two times. What is the probability of two heads?
2. We flip a coin two times. Suppose that the first coin is biased and the probability to flip a heads is $p_1 \in [0,1]$ and the second coin is biased with probability $p_2 \in [0,1]$ of flipping a heads. What is the probability of two heads?

Solution 2.8. Independence allows us to build very complicated probability spaces of many trials by adding coordinates in the natural way. This will be illustrated below.

Part 1. We define $\Omega_1 = \Omega_2 = \{H, T\}$ to be the probability space corresponding to one flip of the first and second coin respectively so that $\Omega = \Omega_1 \times \Omega_2$. Notice that by independence $\mathbb{P}(HH) = \mathbb{P}_1(H) \times \mathbb{P}_2(H)$, which coincides with our natural way to measure the areas of shapes.

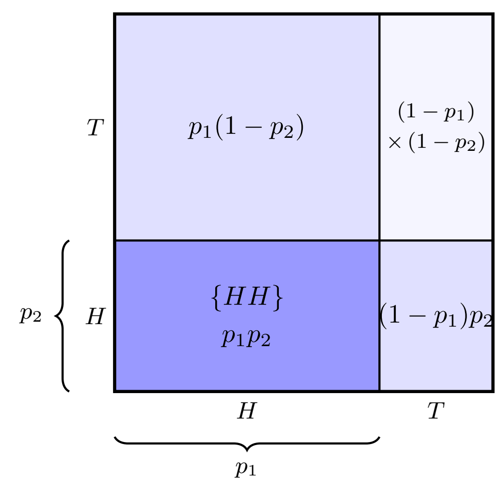


$$\mathbb{P}(HH) = \mathbb{P}_1(H) \cdot \mathbb{P}_2(H) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}$$

Notice that the result coincides with our intuition because $\Omega = \{HH, TH, HT, TT\}$ and all outcomes are equally likely as illustrated above by the areas.

Part 2. Using the same set up as before, by independence

$$\mathbb{P}(HH) = \mathbb{P}_1(H) \cdot \mathbb{P}_2(H) = p_1 p_2.$$



$$\text{total area} = p_1 p_2 + p_1(1-p_2) + (1-p_1)p_2 + (1-p_1)(1-p_2) = 1$$

2.5 Proofs of Key Results

Problem 2.9. Suppose that A and B are independent and mutually exclusive. Show that $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$.

Solution 2.9. On one hand, if A and B are independent, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

On the other hand, if A and B are exclusive

$$\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0.$$

For both to hold simultaneously, we must have $\mathbb{P}(A)\,\mathbb{P}(B) = 0$, so either $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$.

Problem 2.10. (Theorem 1) Prove the inclusion exclusion principle.

Solution 2.10. Notice that A and $(B \setminus A)$ are exclusive events such that $A \cup (B \setminus A) = A \cup B$, so by countable additvity

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$$

Next, notice that $B \setminus A$ and $A \cap B$ are exclusive and $(A \cap B) \cup (B \setminus A) = A \cap B$,

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) \implies \mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

which implies our result. The general statement follows from a straightforward but tedious induction argument. We have

$$\begin{aligned}\mathbb{P}(\cup_{i=1}^{n} A_i) = \mathbb{P}((\cup_{i=1}^{n-1} A_i) \cup A_n) &= \mathbb{P}(\cup_{i=1}^{n-1} A_i) + \mathbb{P}(A_n) - \mathbb{P}((\cup_{i=1}^{n-1} A_i) \cap A_n) \\ &= \mathbb{P}(\cup_{i=1}^{n-1} A_i) + \mathbb{P}(A_n) - \mathbb{P}(\cup_{i=1}^{n-1} (A_i \cap A_n))\end{aligned}$$

The formula is now expressed as the union of up to $n-1$ sets, so we can simplify using the inductive hypothesis to get our result.

Problem 2.11. (Theorem 2) Prove the chain rule for conditional probabilities.

Solution 2.11. By the definition of conditional probabilities,

$$\begin{aligned}\mathbb{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2})\,\mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}) &= \frac{\mathbb{P}(A_1 \cap \cdots \cap A_{n-1})}{\mathbb{P}(A_1 \cap \cdots \cap A_{n-2})} \cdot \frac{\mathbb{P}(A_1 \cap \cdots \cap A_n)}{\mathbb{P}(A_1 \cap \cdots \cap A_{n-1})} \\ &= \mathbb{P}(A_n \cap A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2}).\end{aligned}$$

The exact same computation implies that

$$\mathbb{P}(A_n \cap A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2})\,\mathbb{P}(A_{n-2} \mid A_1 \cap \cdots \cap A_{n-3}) = \mathbb{P}(A_n \cap A_{n-1} \cap A_{n-2} \mid A_1 \cap \cdots \cap A_{n-3}).$$

Each time we do this computation, we can move the right most set on the right of the $\mid$ to the left, so continuing inductively by computing the right most products implies our result.

Problem 2.12. (Theorem 3) Prove the law of total probability.

Solution 2.12. Since $B_1, \dots B_k$ partition the sample space, we have

$$A = A \cap S = A \cap (\cup_{j=1}^n B_j) = \cup_{j=1}^n (A \cap B_j)$$

and the events $\{(A \cap B_1), \dots, (A \cap B_n)\}$ are mutually exclusive because the B_j are. Therefore, by countable additivity,

$$\mathbb{P}(A) = \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_2) + \cdots + \mathbb{P}(A \cap B_n).$$

To prove the second equality, we can use the the chain rule to conclude that $\mathbb{P}(A \cap B_j) = \mathbb{P}(A|B_j)\,\mathbb{P}(B_j)$ for all j, so

$$\mathbb{P}(A) = \mathbb{P}(A \mid B_1)\,\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2) P(B_2) + \cdots + \mathbb{P}(A \mid B_n)\,\mathbb{P}(B_n).$$

Problem 2.13. (Theorem 4) Prove Bayes rule.

Solution 2.13. We start from the definition of conditional probability,

$$\mathbb{P}(B_i \mid A) = \frac{\mathbb{P}(B_i \cap A)}{\mathbb{P}(A)}.$$

By the chain rule, $\mathbb{P}(B_i \cap A) = \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$, so

$$\mathbb{P}(B_i \mid A) = \frac{\mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}{\mathbb{P}(A)}.$$

To get the second equality, the law of total probability states that

$$\mathbb{P}(A) = \sum_{j=1}^n \mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)$$

Problem 2.14. (Corollary 2) Prove the Union bound.

Solution 2.14. This is called the *union bound*. By the inclusion exclusion principle, we see that

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

since $\mathbb{P}(A \cap B) \geq 0$. The general statement then follows by (strong) induction. We have

$$\begin{aligned} \mathbb{P}(A_1 \cup A_2 \cup A_{n-1} \cup A_n) &= \mathbb{P}((A_1 \cup A_2 \cup A_{n-1}) \cup A_n) \\ &\leq \mathbb{P}(A_1 \cup A_2 \cup A_{n-1}) + \mathbb{P}(A_n) \\ &\leq \sum_{k=1}^{n-1} \mathbb{P}(A_k) + \mathbb{P}(A_n) = \sum_{k=1}^n \mathbb{P}(A_k). \end{aligned}$$

Remark 9. It is true that equality is attained if $A_1, \dots, A_n$ are mutually exclusive. However, we can have equality even if $A_1, \dots, A_n$ are not mutually exclusive. We can consider the distribution on the two points $\{a, b\}$ such that $\mathbb{P}(\{a\}) = 0$ and $\mathbb{P}(\{b\}) = 1$. Then $A = \{a, b\}$ and $B = \{a\}$ satisfy the union bound, but they are not mutually exclusive.